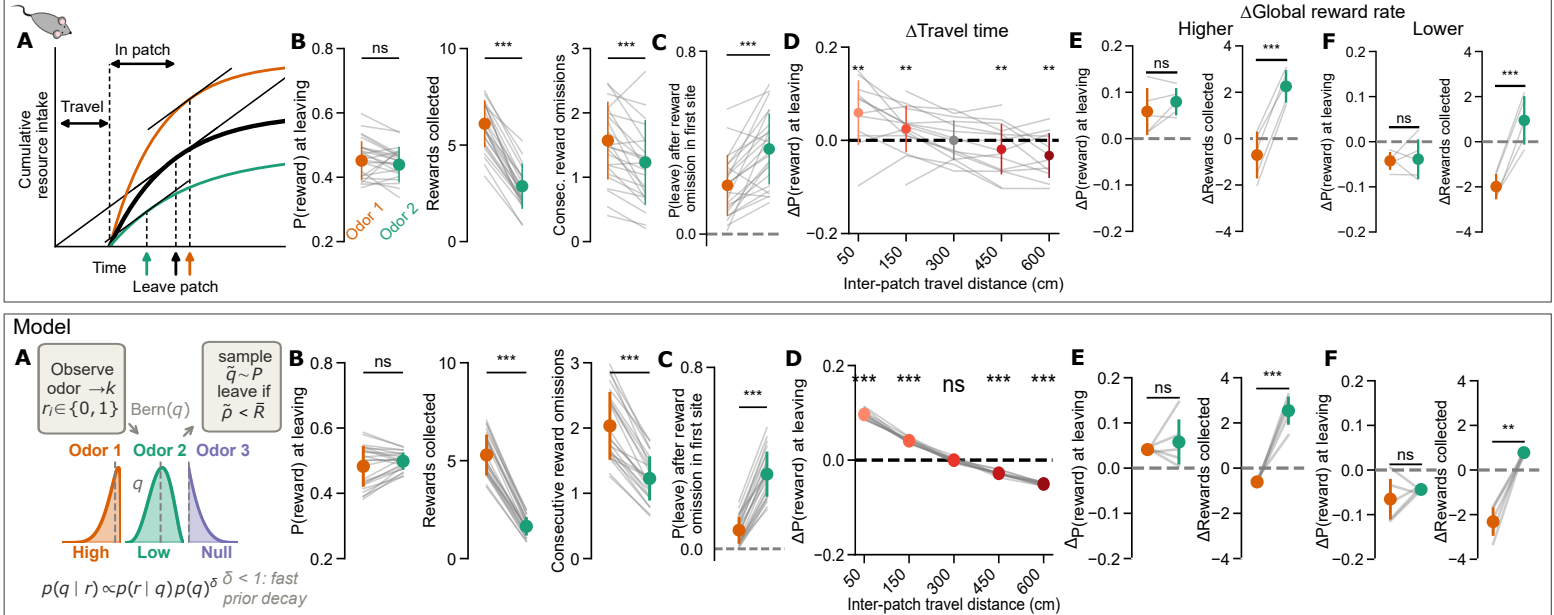

A
In patch
Travel
Cumulative resource intake
Time
Leave patch
B
ns
***
P(reward) at leaving
Odor 1
Odor 2
Rewards collected
Consec. reward omissions
C
P(leave) after reward omission in first site
D
ΔTravel time
ΔP(reward) at leaving
Inter-patch travel distance (cm)
ΔGlobal reward rate
Higher
Lower
E
F
ΔRewards collected
Model
Observe odor →k
r_i∈{0,1}
Bern(q)
sample q̃∼P
leave if p̃ < R̄
Odor 1
Odor 2
Odor 3
High
Low
Null
p(q | r) ∝ p(r | q) p(q)^δ
δ < 1: fast prior decay
Consecutive reward omissions